

Home Automation System

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In this era of digital revolution, we are surrounded by smart devices that are capable of making decisions on their own without much human intervention. Our home can also be made smart by implementing a real-time home automation system that monitors parameters like power consumption and human presence.

Home automation may include centralised control of electrical devices including lightings, appliances and security.

Presented here is a touch-control based home automation system that can control up to six electrical devices. It also has a separate keyboard interface module for troubleshooting and system settings.

This system consists of closely-

networked Atmel's ATmega8, which is an AVR based microcontroller with 512 bytes EEPROM on a 28-pin DIP package, 1024 bytes internal SRAM and 8kB internal flash memory. The complete system is assembled in a small, portable central processing unit (CPU) chassis for an aesthetic look and uninterrupted 24x7 usage. Author's prototype is shown in Fig. 1.

Circuit and working

The circuit has four sections: main module, relay module, touch-control module and keyboard interface module. The system is housed on the CPU chassis of a desktop computer and is powered by 400W ATX power supply for error-free operation and proper power delivery to the system.

Advanced Technology eXtended (ATX) is a motherboard form factor specification developed by Intel in 1995 to maintain the integrity and sustainability of the system.

Main module. The circuit of the main module is shown in Fig. 2. It has ATmega8 microcontroller,



Fig. 1: Author's prototype

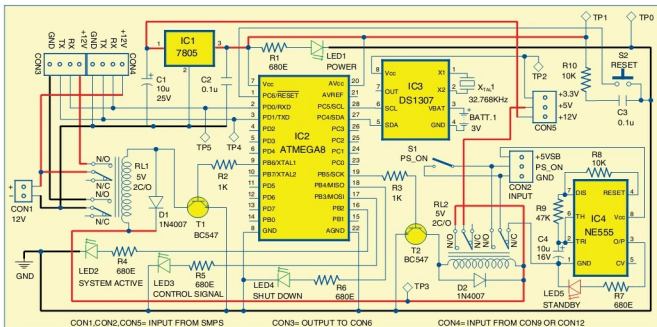


Fig. 2: Circuit of the main module